

HelixCode — CONTINUATION

CONTINUATION — HelixCode web-IDE platform

Revision: 8 · **Updated:** 2026-07-02 · **Status:** codeserver-1.0.0-dev-0.0.4 **RELEASED** (0e2f854, tag on all 4 mirrors, GitHub+GitLab prereleases). Release gate: full §11.4.169 matrix **29/29 suites PASS, 0 FAIL** (61 checks PASS, 24 honest SKIP; evidence qa-results/run_all/20260701T222753Z-2813294). **Post-release coverage promotion (§11.4.52/§11.4.123):** wiring an authorized HELIX_TEST_SSH_KEY (gitignored deploy/.env; §11.4.10 never committed/logged) promoted the credential-gated auth SKIPS into **real authenticated-session PASSes**, and a non-destructive **isolated throwaway-gate** C3 variant (§11.4.85/§11.4.101 — own loopback port, throwaway keypair/secret, no live-gate rotation, no real credential) closed the LAST credential_absent SKIP — matrix **29/29 suites PASS, 0 FAIL, 72 checks PASS / 16 SKIP, ZERO credential_absent** (evidence qa-results/run_all/20260702T000047Z-3718172); the 16 remaining SKIPS are all honest (13 topology_unsupported = 11 legacy-suite retirements + 2 Chromium-contended pixel proofs, 2 operator_attended with passing autonomous CLI siblings, 1 feature_disabled_by_config = load_auth L3 login-flood rate-limit inconclusive, where the limiter is positively confirmed by security_auth #4's real 429). C3's §1.1 pair binds: it PASSes on the real gate (200/200/401/200) and FAILs on a gate with the cookie HMAC check disabled. Ships: login-redirect fix + login-form copy/paste clipboard buttons + Open VSX marketplace install/use/persist/config + popular extensions + VS Code Dark default theme (host-rendered pixel proof) + durable systemd --user edge. Three test-infra suites hardened against shared-host ulimit -u fork-pressure false-FAILs (§11.4.1/§11.4.3/§11.4.174) + independently reviewed to a clean GO (§11.4.142/§11.4.134). Stack LIVE at <https://192.168.0.213:52443> (all systemd --user).

Read FIRST on any fresh session: this file, then `git fetch --all`, then the auth-pivot spec + AUTH guide + feature ledger below. This is the §12.10 / §11.4.131 standing resumption anchor.

Current phase — codeserver-1.0.0-dev-0.0.4 **RELEASED** (idle unless new operator request)

codeserver-1.0.0-dev-0.0.4 is **RELEASED** (0e2f854): full §11.4.169 release-gate matrix **29/29 suites PASS, 0 FAIL** on the reviewed+hardened tree (evidence qa-results/run_all/20260701T222753Z-2813294), tag pushed to all 4 mirrors (no force, §11.4.113), GitHub + GitLab prereleases created. The batch's three hardened test-infra suites (race, extensions_ui_auth, extensions_popular_auth) were fixed for shared-host ulimit -u fork-pressure **environmental** false-FAILs (§11.4.1/§11.4.3/§11.4.174 — genuine defects still FAIL, host starvation → SKIP-with-reason) and independently reviewed to a clean GO (§11.4.142/§11.4.134; F1-F4 + N1 all resolved; every §1.1 mutation re-proven).

Prior: codeserver-1.0.0-dev-0.0.3 **RELEASED** (2746e0e) — the real-account SSH-key challenge-response auth model, §11.4.169 matrix 23/23 + Go gate 70 tests -race. The dev-0.0.4 batch (login-redirect fix + copy/paste buttons + Open VSX marketplace coverage + VS Code Dark default + durable edge + deep research) landed across these commits toward the release:

- 75e2d9b — login-redirect fix (an unauthenticated browser now lands on the login form instead of a bodyless “This page isn’t working”).
- e0519bf — hardening: **DURABLE** systemd Quadlet edge that survives session crashes + full-host reboot, tarball code-server install (drops the fragile npm path), loopback:8080 host-firewall rule, doc exports.
- 43e37a5 — extensions_auth suite **5/5** (Open VSX plugin install + use, anti-bluff).
- eee6169 — feature ledger reconciled to real evidence (docs/features/Status.md
 - Status_Summary.md).
- 849dc9a — extensions (marketplace) guide + read-only gallery probe.
- 8351e4c — login-form copy/paste clipboard buttons **3/3** + UI-driven marketplace test.
- 97a8c69 — VS Code Dark **default** theme (+ pixel proof) + popular-extensions install/use/persist + auth-modernization deep-research report.

The stack is LIVE + reachable at **https://192.168.0.213:52443** — every component is systemd --user-managed (Caddy edge Quadlet, helix-auth, helix-code-server). All standing operator

requests have landed: extension install/use tests, login-form copy/paste buttons, popular-extensions install/use/persist, VS Code Dark default, and the first deep-research report.

- Changelog (in progress): docs/changelogs/codeserver-1.0.0-dev-0.0.4.md
- Prior release: docs/changelogs/codeserver-1.0.0-dev-0.0.3.md
- Auth pivot spec (authoritative): docs/superpowers/specs/2026-07-01-auth-pivot-ssh-key.md
- Extensions guide: docs/guides/EXTENSIONS.md · Edge boot: docs/guides/EDGE_BOOT.md
- User guide: docs/guides/AUTH.md · Feature ledger: docs/features/Status.md
- Deep-research (standing directive): docs/research/auth_modernization_20260701/FINDINGS.md (first report → **WebAuthn/passkeys recommended**)

Why the pivot (captured facts, §11.4.6)

This host is ALT Linux with the **tcb** shadow scheme, so a **non-root** service cannot verify passwords via PAM (/etc/tcb/<user>/shadow unreadable by the user; tcb_chkpwd setuid helper permission-denied to non-root ⇒ pam_authenticate returns PAM_AUTH_ERR for every password). A password gate would need root; the operator directive is **no sudo/root** and **all access via SSH keys**. ~/.ssh/authorized_keys is readable and ssh-keygen -Y sign/-Y verify are supported, so SSH-key challenge-response runs fully **as non-root milosvasic**, with **no password and nothing stored**.

Architecture (this release)

Browser —HTTPS→ Caddy (TLS edge, CONTAINERIZED rootless Podman; HTTP/3 + brotli via custom xcaddy)
└─ forward_auth → helix-auth (host-native Gin, systemd --user, loopback:8081, NON-root)
└─ reverse_proxy → code-server (host-native AS milosvasic, systemd --user, --auth none, loopback:8080; inherits ~/.ssh, ~/.bashrc, host binaries)

- **code-server** — host-native systemd --user, --auth none, loopback:8080.
- **helix-auth** — host-native **Gin** gate, systemd --user, loopback:8081, non-root: SSH-key challenge (GET /login) → verify pasted signature vs authorized_keys (ssh-keygen -Y verify) → __Host-session cookie; GET /auth for forward_auth; **fails closed**.
- **Caddy** — sole containerized component: TLS edge, forward_auth (fail-closed), HTTP/3, encode zstd br gzip, /proxy 403, X-Helix-User strip.

- **Stack directive adopted:** Go + Gin + HTTP/3 (QUIC) + Brotli.

Feature status (§11.4.6 — honest)

The real-account + SSH-key rows in docs/features/Status.md are **PASS with cited captured evidence** (§11.4.169 matrix 23/23 PASS; Status.md Revision 2). Testing: Go unit/integration/race (**70 tests, -race, 81.8% cover**) + shell suites tests/types/{e2e,security,stress_chaos,concurrency,load,memory,benchmark,challenges,helixqa}_auth.sh + banks tests/banks/helixcode-auth-{challenges,helixqa}.yaml. The live aggregate is the conductor-filled placeholder in the changelog.

Honest boundaries carried into the release

- Editor “jail” is **cosmetic** (Explorer default only; terminal/Open-Folder/ extensions keep full host access by design).
- --auth none on loopback:8080 is a **residual** — host firewall rule for loopback:8080 recommended (all real auth is at the gate/edge).
- code-server pinned **4.117.0** (newest on npm at release time).
- Real public Let’s Encrypt remains **operator-gated** (see docs/guides/TLS.md).

Migration

scripts/install-auth.sh (host-native, **no sudo**) provisions the systemd --user code-server + helix-auth units; deploy/up.sh brings up the Caddy edge. **CODE_SERVER_PASSWORD is retired** → HELIX_AUTH_MODE=sshkey, HELIX_AUTH_ACCOUNT, HELIX_AUTH_AUTHORIZED_KEYS, HELIX_AUTH_PRINCIPAL, PROJECTS_ROOT (no password parameter).

Immediate NEXT (post-release — idle per §11.4.126 unless a new operator request)

codeserver-1.0.0-dev-0.0.4 is RELEASED + published everywhere; the post-release credential-coverage promotion (above) is landed and matrix-proven. Open follow-ups (non-blocking, no new operator request required to be idle):

1. **C3 throwaway-gate variant — DONE** (isolated helix-auth on its own loopback port + throwaway keypair/secret; stress_chaos_auth C3 now runs autonomously + non-

destructively, closing the last `credential_absent` SKIP; §1.1 pair binds). Fixture helpers `hc_spawn/restart/stop_throwaway_gate` in `tests/lib/stack_fixture.sh`.

2. **(Optional) harness auto-resolve of an authorized key** — make real auth coverage the out-of-the-box default even without `deploy/.env` wiring (scan `~/.ssh` for a key whose `.pub` is in the gate's `authorized_keys`; honest SKIP if none). Needs the full review gauntlet + a paired §1.1 mutation; not required (the `deploy/.env` wiring already closes the gap durably on this host).
3. Standing operator directive — **frequent deep research** (§11.4.150): keep producing reports; the first landed report recommends **WebAuthn/passkeys** as the auth-modernization path (`docs/research/auth_modernization_20260701/FINDINGS.md`); the passkey-login spec at `docs/superpowers/specs/2026-07-01-webauthn-passkey-login.md` is drafted, NOT implemented, awaiting an operator decision.

Binding constraints (every phase)

Port band **52000-52999**; publish only edge ports on `0.0.0.0`; **no secrets** (only git-ignored `.env`); **every gate paired with a §1.1 mutation**; **no --force/--no-verify/bypass**; hardlinked `.git` backup before destructive ops; **anti-bluff** (PASS needs captured runtime evidence, no invented numbers); stop + root-cause + full retest on any defect; release prefix codeserver; submodule commits propagate first; docs kept in sync.

Prior state (Phases 1-2, retained)

Phase 1 COMPLETE; Phase 2 core working+verified (Caddy edge + code-server stack on Podman; TLS1.3 edge on 52443, →301 on 52080; \$PROJECTS bind-mounted). Owned submodules (8) at root carry the Constitution inheritance pointer; constitution pinned (see history). Prior release `codeserver-1.0.0-dev-0.0.2` shipped Let's Encrypt HTTPS (auto-renew + rotation) + the 14-suite §11.4.169 matrix.